

EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

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Abstract

001 Reinforcement fine-tuning of vision-language navigation
002 (VLN) policies commonly relies on destination suc-
003 cess or similarity to a reference route. These signals
004 leave an instruction’s intermediate requirements unre-
005 solved: reaching the endpoint does not establish that
006 the agent entered the requested room or passed the
007 specified landmark. We study an online reward that
008 represents an instruction as an ordered list of seman-
009 tic events and verifies each event against observations
010 before and after executed actions. A frozen vision-
011 language model supplies a three-way verdict (complete,
012 incomplete, uncertain); motion and stopping checks re-
013 ject visually plausible but unexecuted events. Each
014 event is rewarded at most once, and the process re-
015 ward is bounded below the success bonus. We inte-
016 grate EventTrace into multi-turn ActiveVLN/GRPO
017 and compare it with destination-only training for 64
018 optimization steps across three matched seeds. All six
019 checkpoints and the SFT initialization are evaluated
020 on the same 1,839 R2R val-unseen episodes. Event-
021 Trace changes success rate by +0.65, +0.44, and -2.88
022 percentage points across seeds; its mean paired change
023 is -0.60 points (sample standard deviation 1.98). A
024 single-AI blind review of 49 selected transitions finds
025 that 7 of 13 verifier completion predictions match its
026 labels, without independent human adjudication. The
027 method is operational, but these results do not estab-
028 lish a navigation improvement or reliable semantic ver-
029 ification.

030 1. Introduction

031 Vision-language navigation asks an embodied agent to
032 follow a natural-language route through visual obser-
033 vations. The Room-to-Room (R2R) task introduced
034 instruction-guided navigation in real indoor environ-
035 ments [1]; VLN-CE moves this task into continuous

control [5]. Success rate (SR) and success weighted by
path length (SPL) capture destination attainment, yet
a successful stop can coexist with skipped intermedi-
ate instructions. Instruction fidelity work made this
gap explicit [4].

Recent reinforcement fine-tuning (RFT) approaches
give visual-language policies online or trajectory-level
feedback. ActiveVLN performs multi-turn exploration
with group-relative policy optimization (GRPO) [8,
10]. VLN-R1 uses time-decayed action rewards [7];
Nav-R1 combines format, understanding, and naviga-
tion rewards [6]; SeeNav-Agent proposes step-level ver-
ifiable process rewards in another navigation bench-
mark [9]. Our question is narrower: can a naviga-
tion instruction’s ordered, physical events provide a
bounded reward during closed-loop R2R-CE rollouts,
without matching the policy’s actions to a reference
path?

We introduce EventTrace, a modular reward on top
of an existing multi-turn policy. The reward uses a
short event list, a frozen visual verifier with an absten-
tion state, deterministic checks of motion or stopping,
and a once-only order-preserving tracker. The actor
sees only the original egocentric observations and in-
struction; simulator displacement is available only to
the reward. Figure 1 illustrates the distinction between
seeing a bathroom and actually entering it.

We evaluate the reward with matched three-seed
training and complete R2R val-unseen coverage. The
observed seed variation and an exploratory verifier au-
dit limit the empirical claim: this implementation is
reproducible but does not improve navigation consis-
tently.

2. Related Work

Path and instruction fidelity. SR and SPL reward the
destination and efficiency, whereas nDTW and cover-
age weighted by length score (CLS) compare trajec-
tories with a reference route [3, 4]. Reference trajec-

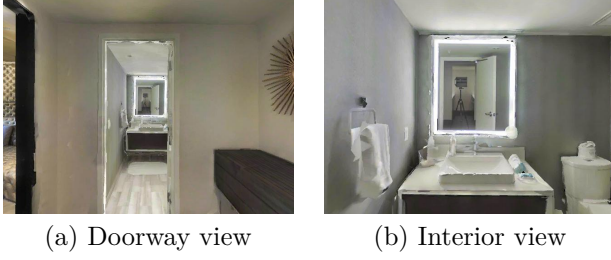


Figure 1. Real simulator frames from a diagnostic replay of R2R training episode 2889. Visibility alone does not complete the event “enter bathroom.” The verifier uses before/after views and executed motion; the ordered tracker later checks “stop near sink.” These frames illustrate the mechanism and are not an independent accuracy test.

ries can be useful supervision but do not directly answer whether a distinct visual event actually occurred when multiple valid routes exist. Our reward targets event completion during execution; route metrics remain complementary evaluation measures.

RL for VLN. ActiveVLN is our implementation base because its environment can return feedback after each multi-action turn [10]. VLN-R1 and Nav-R1 show other ways to shape VLN rewards [6, 7]. SeeNav-Agent is especially relevant in its emphasis on step-level verification [9]. EventTrace differs in its ordered event state, explicit uncertainty outcome, and physical gates against rewarding a landmark seen without the requested transition. Our pilot still applies GRPO to trajectory-level returns; it does not claim SeeNav-Agent-style step-level credit assignment.

3. Method

3.1. Task and event representation

An instruction x yields a list $E(x) = (e_1, \dots, e_K)$, where e_k contains a verb type, target phrase, and source span. Supported event types include enter, pass, descend, and stop_near. A frozen vision-language model (VLM) proposes at most three events, restricted to source spans in the instruction. The eight-episode wiring pilot uses curated lists; the main 64-step training runs parse instructions automatically at episode start. Parser errors produce an empty list and no semantic reward. We have not independently audited parser accuracy.

In the curated pilot, each of the eight episodes contains a stopping event; there are also three enter, three pass, and three descend events, for 17 in total. These counts describe the pilot distribution, not the prevalence of event types in R2R. Table 1 shows the deter-

Table 1. Motion gates in the pilot. The VLM must also verify that the image pair supports the specified target.

Event	Count	Required gate
enter	3	Forward action; $\ \Delta p\ \geq 0.10$ m
pass	3	Forward action; $\ \Delta p\ \geq 0.10$ m
descend	3	Vertical change $\Delta y \leq -0.08$ m
stop_near	8	Executed STOP action

ministic portion of the verifier. Each gate can reject completion before a VLM call, but passing a gate is never sufficient to award a point.

At turn t , the policy receives the instruction, action history, and current egocentric image I_t , and emits up to three low-level actions. The simulator executes these actions and returns I_{t+1} , pose change Δp_t , and task metrics. Only the current pending event e_{k_t} is queried. The frozen verifier produces

$$v_t = V(e_{k_t}, I_t, I_{t+1}, a_t, \Delta p_t) \in \{Y, N, U\}, \quad (1)$$

where U denotes insufficient evidence. A view of a landmark is not by itself a completion verdict. For crossing and passing, a forward displacement check is required; descending requires a negative vertical displacement; stopping near a target requires an executed STOP. The VLM inspects whether the image pair supports the specified target and transition. These are deliberately conservative gates, not a learned proof of physical state.

3.2. Ordered, bounded process reward

Let $k_t \in \{1, \dots, K+1\}$ index the first unfinished event. The event fires only when its verifier says Y. Then

$$c_t = \mathbb{I}[k_t \leq K \wedge v_t = Y], \quad (2)$$

$$k_{t+1} = k_t + c_t, \quad r_t^{\text{sem}} = \frac{\lambda}{K} c_t. \quad (3)$$

A N or U verdict leaves k_t unchanged. The tracker prevents repeated reward for the same landmark, enforces event order, and guarantees $\sum_t r_t^{\text{sem}} \leq \lambda$. Because only the pending event is checked, at most one event fires per turn even if the agent performs several actions. This avoids ambiguous credit when a model emits an action batch but can delay recognition of adjacent events. The tracker retains a confirmed event after later image changes, so a completed event cannot be unconfirmed by viewpoint drift.

The base destination term is the weighted-success reward already used by ActiveVLN, with nDTW weight zero. We add a fixed success floor b to both arms:

$$R(\tau) = \sum_t r_t^{\text{sem}} + S(\tau) \left[b + \alpha \left(1 - \min \left(\frac{d_T}{3}, 1 \right) \right) \right], \quad (4)$$

where $S(\tau)$ is simulator success after STOP, d_T is terminal distance in meters, $\alpha = 15$, $b = 2$, and $\lambda \in \{0, 1\}$. Hence any success receives at least 2 while a failure receives at most 1 semantic point. This priority holds for this nonnegative reward configuration; it does not imply that the verifier is correct. The bound also makes failure reward interpretable: partial event completion can distinguish two unsuccessful attempts without allowing a failed route to outrank a successful one solely through event bonuses. The STOP event is distinct from simulator success: it demands visual proximity to its instruction target, while $S(\tau)$ uses the simulator’s goal region. Disagreement between these signals is precisely a case the proposed reward aims to expose.

For each training episode, we sample two trajectories and normalize their returns within the group for GRPO [8]. Event rewards are emitted after each turn, but the current trainer sums them before constructing a trajectory advantage. Consequently, the implementation changes which trajectories are preferred, not which individual turn receives the policy gradient. Turn-level event timing is available in the log for future credit-assignment work.

3.3. Verifier failure policy

The verifier is a frozen Qwen3-VL-8B-Instruct model. A service timeout, malformed response, or uncertain decision yields zero event reward; it does not end the episode. The reward requires no reference action prefix. Simulator pose is used only for reward verification, never added to the actor observation. In the pilot, the fixed event manifest is matched to both episode ID and exact instruction text to avoid assigning another episode’s event list; the main runs use online parsing. The verifier sees two RGB frames resized to fit within 448×448 pixels, the event type and target, its source phrase, the executed action text, translation magnitude, and vertical displacement. Its output is a deterministic one-letter Y/N/U response; generated explanations are not treated as confidence scores. This makes the reward externally auditable through a saved per-turn verdict, although a forced-choice label is still fallible. The HTTP verifier is separated from the simulator so it can be frozen and logged independently of policy updates.

4. Experimental Setup

Environment and policy. We use R2R train in VLN-CE, rendered from Matterport3D scans [2], the ActiveVLN implementation, and an SFT-initialized Qwen2.5-VL-3B policy. Matterport3D data and derived imagery are subject to its [academic terms of use](#). Each turn contains at most three actions; the limit is

12 turns and 36 simulator steps. Actor sampling uses temperature 1.2 and two independently sampled rollouts per item. GRPO trains on two A800 GPUs at learning rate 10^{-6} with batches of four episodes. The destination-only control sets $\lambda = 0$; the event arm sets $\lambda = 1$. Both arms have the same SFT initialization, destination reward, data order within seed, and optimizer budget.

Pilot and corrected main training. We first replay eight training episodes with expert actions solely to inspect expected events, then run a two-step reward-wiring pilot with curated events. These replays are neither held-out labels nor training results for the main comparison. For the main study, each arm runs 64 optimization steps at seeds 11, 22, and 33: 256 train-episode exposures and 512 sampled rollouts per arm and seed. The event arm uses automatically parsed events. A preflight check detected that a fixed per-request sampling seed duplicated every two-rollout GRPO group and yielded zero actor gradients; we excluded that attempt, removed the per-request seed while retaining the engine seed, and reran all six models. Every corrected group has distinct sampled actions. The archive records the excluded attempt and corrected patch.

Full val-unseen evaluation. SR is the fraction of trajectories satisfying the simulator goal condition; SPL also accounts for path efficiency. Mean terminal goal distance is in meters. Training rollout success and return are diagnostics, not held-out outcomes; returns differ in definition between arms. The SFT model and six corrected step-64 checkpoints run on the identical manifest of all 1,839 R2R val-unseen episodes from 11 scenes. We use one generation per episode, temperature 0.2, top- p 0.8, a 12-turn limit, and at most 76,800 image pixels. Four Habitat shards per checkpoint divide the manifest without overlap. The evaluator uses the same prompt and environment-specific OpenAI-compatible request and image-resize shim for all seven models. The completion gate checks the exact episode-ID set and zero inference errors. Episode IDs are paired within each seed, but stochastic decoding uses only one generation per checkpoint and is an additional source of variation.

Blind verifier review. Separately, we collect shortest-path follower trajectories from 16 different val-unseen episodes, of which 12 yield at least one automatically parsed event. Before querying the verifier, one AI-assisted reviewer inspects RGB image pairs, executed actions, and motion and fixes 49 three-way completion

Table 2. Observed pilot results on eight training episodes. Each arm ran two optimizer steps and produced 16 sampled rollouts. The event arm’s larger mean return is expected from the added reward and does not indicate better navigation.

Metric	Destination only	EventTrace
Successful rollouts	13/16	13/16
Mean trajectory return	9.325	10.455
Mean event progress	–	0.656
Confirmed event hits	–	23
Uncertain verifier turns	–	33
Verifier service errors	–	0

labels across those 12 episodes. The sample includes apparent transitions, controls, and final stops. Labels are independent of verifier predictions, but there is no second reviewer or human ground-truth adjudication, and the selected class proportions are not representative of deployment traffic. We therefore treat this as a failure analysis, not a population accuracy estimate.

5. Results and Analysis

Reward reaches optimization. In the first optimizer step, both arms sample the same eight action sequences and attain seven successes. For every matched trajectory, the difference in total return equals its semantic event score, including a failed episode 7110 trajectory that receives 0.5 for completing “descend stairs” but never reaches the destination. The modified return is consumed by the training reward manager, and both arms save step-2 checkpoints after nonzero policy updates. This verifies the wiring of Eq. 4; it is not evidence that event reward improves the policy.

What the verifier recognizes. In diagnostic expert replays, four of eight episodes receive full event credit, two receive partial credit, and two receive none; the tracker confirms 11 of 17 curated events. For episode 2889, seeing the sink from outside the bathroom produces uncertain verdicts before entrance, while later motion and view changes confirm enter. A rotation followed by premature STOP yields zero semantic credit in a negative replay. There are also false negatives: one shallow stair descent is rejected by the vertical-motion threshold, and one landmark is never confirmed. Because the events were curated using these episodes and the replays lack independent annotations, 11/17 must not be reported as verifier recall or accuracy.

Main training diagnostics. All six corrected runs finish 64 steps, with 256 diverse GRPO groups per arm and seed. Training-rollout success is 111/512 versus

Table 3. Complete R2R val-unseen evaluation. Every row uses the same 1,839 episodes from 11 scenes, one generation per episode, and zero inference errors. Goal distance is in meters. C: destination-only control; E: EventTrace.

Checkpoint	Successes	SR (%)	SPL (%)	Dist.
SFT	544	29.58	28.69	7.03
Seed 11 C	479	26.05	25.32	7.75
Seed 11 E	491	26.70	25.91	7.63
Seed 22 C	566	30.78	30.14	6.91
Seed 22 E	574	31.21	30.63	6.98
Seed 33 C	502	27.30	26.65	7.14
Seed 33 E	449	24.42	23.16	7.76

119/512 (control versus event) for seed 11, 108/512 versus 119/512 for seed 22, and 88/512 versus 82/512 for seed 33. These on-policy samples change as training proceeds, so they are not comparable to the held-out evaluation. The event verifier records 272, 275, and 243 confirmed event hits over the three event-arm runs. These counts are not event accuracy: events are automatically parsed and no independent labels cover these rollouts.

Complete val-unseen results. Table 3 gives all seven checkpoints on the same 1,839 episodes. Within matched seeds, EventTrace changes SR by +0.65, +0.44, and −2.88 percentage points, and SPL by +0.58, +0.49, and −3.49 points. The mean paired seed change is -0.60 ± 1.98 points for SR and -0.80 ± 2.32 points for SPL (mean $\pm$ sample standard deviation over three seeds). At the episode level, event-only versus control-only successes are 157 versus 145, 197 versus 189, and 155 versus 208. This variation does not support a consistent improvement. All seven evaluations have exact 1,839-episode coverage and zero logged inference errors. The earlier 16-episode, two-step probe yielded 6/16 successes for every arm; it was exploratory and is superseded by the complete comparison.

Verifier failure analysis. Table 4 compares the frozen verifier with labels fixed before its predictions were obtained. Seven of eight reviewed completions receive Y, but only 7 of 13 Y predictions match a reviewed completion; two are on clear non-completions and four on ambiguous turns. One false positive occurs after the agent has already entered the requested room, exposing a failure to distinguish current state from an event newly completed this turn. Motion gates alone do not eliminate this error. These selected examples cannot establish deployment precision, but they make verifier reliability a concrete limitation of the present reward.

Table 4. Blind review of 49 selected val-unseen event transitions from 12 episodes. Rows are review labels and columns are frozen verifier predictions. U means evidence insufficient; the review was AI-assisted and has no independent human adjudication.

	Pred. Y	Pred. N	Pred. U
Review Y	7	0	1
Review N	2	15	12
Review U	4	1	7

6. Discussion and Limitations

The main risk is reward misspecification. A VLM may mistake a visually similar doorway for the requested one, and hard displacement thresholds may miss legitimate small movements. Ordered one-at-a-time verification can also miss rapid completion of two events in one turn. The verifier receives before/after frames rather than a continuous video; event timestamps are only turn precise. Finally, a trajectory-level GRPO advantage offers limited credit assignment to the action that completed the event.

The three-seed, complete val-unseen comparison reveals a negative mean effect and large seed variation, but 64 steps expose only 256 training-episode instances per run. Stochastic evaluation uses one decoding sample per episode, and we have not tested a larger training budget or repeated decoding. No independent human adjudication of verifier transitions is available; the 49-example blind review has only one AI labeler and was enriched for events. Automatic instruction parsing is also unaudited. Thus we cannot attribute the negative result uniquely to noisy verification, parser errors, reward scale, training budget, or policy optimization. Human-labeled verifier and parser sets, then ablations of event ordering, abstention, motion gates, reward budget, and turn-level advantage are needed to isolate causes. Event precision or recall cannot responsibly be reported from the current audit.

7. Conclusion

We formulated a path-free semantic process reward for closed-loop VLN that verifies ordered instruction events against executed motion and before/after views. A wiring pilot confirms that the bounded reward enters online GRPO. In three matched 64-step training runs and complete 1,839-episode val-unseen evaluation, EventTrace has a mean paired SR change of -0.60 percentage points relative to destination-only training, with inconsistent seed effects. A single-AI blind review exposes verifier disagreement but provides no in-

dependent ground truth. The current method is reproducible; reliable semantic verification and a navigation benefit remain unproven.

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